

Tracking Performance Drift in LLMs

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User prompt: Generate a scholarly research paper on the topic: "Tracking Performance Drift in Large Language Models Across Successive Version Updates". The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.

Response: Tracking Performance Drift in Large Language Models Across Successive Version Updates

Abstract Large Language Models (LLMs) are increasingly integrated into critical enterprise and academic systems. However, the continuous, often unannounced, updating of proprietary models introduces a phenomenon known as "performance drift" or "behavioral drift." This paper explores the manifestations of temporal degradation and capability shifts in state-of-the-art models like GPT-3.5 and GPT-4. By reviewing contemporary literature, we examine how model outputs change regarding instruction following, code generation, mathematical reasoning, and safety alignment. Furthermore, this paper synthesizes current evaluation frameworks, challenges in maintaining contextual equilibria during multi-turn interactions, and proposes future directions to address the opacity of continuous model deployment. The findings emphasize the necessity for robust, longitudinal monitoring frameworks to ensure the reliability and safety of LLMs in production environments.

Keywords: Large Language Models, Performance Drift, Temporal Degradation, Generative AI, Behavioral Consistency, AI Alignment.

1. Introduction

The widespread adoption of Large Language Models (LLMs) has fundamentally transformed natural language processing, automated reasoning, and decision-support systems. Models such as GPT-3.5 and GPT-4 serve as the backbone for countless commercial applications, ranging from autonomous coding assistants to complex

information retrieval pipelines. However, because proprietary models are served primarily through Application Programming Interfaces (APIs), the underlying architecture, training data, and alignment techniques (such as Reinforcement Learning from Human Feedback) are subject to continuous refinement by their developers.

These silent updates frequently result in what researchers define as *performance drift*—the gradual or sudden shift in a model's responses to identical inputs over time. While some updates successfully mitigate vulnerabilities or improve factuality, they can simultaneously degrade performance in other specialized domains, causing unexpected breakages in downstream applications. This paper provides a comprehensive thematic discussion on how LLMs exhibit behavioral drift across successive versions, highlights the technical challenges in evaluating these shifts, and outlines future avenues for achieving stable, predictable generative models.

2. Background and Literature Review

The study of behavioral consistency in generative AI is a burgeoning field driven by the rapid deployment of commercial LLMs. Historically, software regression testing could rely on deterministic outputs. In contrast, LLMs demonstrate inherent stochasticity, meaning that even with strict temperature parameters, tracking exact behavioral degradation requires statistical frameworks over large query volumes.

Extensive literature has begun to document these shifts empirically. A pivotal study by Chen, Zaharia, and Zou (2024) systematically tracked the behavior of ChatGPT across its March 2023 and June 2023 versions, revealing dramatic fluctuations in its capacity to solve math problems, generate usable code, and navigate sensitive questions. Complementing this, comprehensive surveys have attempted to benchmark the actual power of these models across varied disciplines, noting that capabilities often hinge on opaque training mixtures that change without public documentation (Liu et al., 2024).

Furthermore, behavioral drift affects safety mechanisms and alignment. As models are updated to prevent harmful outputs, their propensity to generalize safely without sacrificing utility can fluctuate. For instance, testing GPT-4 on reliable misinformation mitigation highlights how uncertainty and generalization metrics shift when the model's underlying guardrails are patched over time (Pelrine et al., 2023).

3. Main Thematic and Technical Discussion

Performance drift in LLMs manifests across several distinct axes. The most well-documented areas include mathematical logic, programming proficiency, and multi-turn contextual reasoning.

3.1 Degradation in Step-by-Step Reasoning

One of the most profound observations of performance drift is the alteration in a model's

ability to follow complex reasoning chains. Chen et al. (2024) demonstrated that GPT-4's accuracy on identifying prime versus composite numbers plummeted from 84% in March 2023 to just 51% in June 2023. This regression was largely attributed to a sudden drop in the model's willingness to follow chain-of-thought prompting instructions. Conversely, the same time window saw improvements in GPT-3.5 for identical tasks, illustrating that updates do not yield uniformly positive or negative results across model families.

3.2 Code Generation and Formatting Drifts

LLMs have been aggressively adopted as programming assistants, raising questions about their ultimate viability for complex software engineering (Tian et al., 2023). Tracking models longitudinally reveals that code generation tasks are highly susceptible to formatting drift. Models updated to provide more verbose explanations often begin wrapping code blocks in extraneous Markdown or conversational filler, which breaks automated parsing scripts used by developers. The propensity to inject formatting mistakes has been shown to increase after certain alignment updates, requiring users to continually rewrite their system prompts to maintain consistent syntax (Chen et al., 2024).

3.3 Multi-Turn Context Equilibria

While single-turn tasks highlight immediate regressions, real-world deployment frequently involves multi-turn interactions where context drift becomes a critical failure point. As a conversation elongates, models can deviate from goal-consistent behavior. Recent dynamic frameworks conceptualize this temporal divergence as a stochastic process, showing that multi-turn drift can settle into stable equilibria rather than unbounded degradation, provided that appropriate reminder interventions are implemented to realign the model (Dongre et al., 2025).

4. Current Approaches and Developments

To counteract the unannounced changes in commercial APIs, the research community has developed several methodologies to track and mitigate performance drift.

- **Behavioral Fingerprinting:** Researchers are utilizing standardized suites of queries—acting as a behavioral fingerprint—to regularly probe APIs. By measuring the lexical similarity, unique output fractions, and exact-match accuracy against historical baselines, developers can quantify drift.
- **Prompt Robustness Techniques:** Because models like GPT-4 frequently alter how they interpret specific instructional phrasing over time, engineers are adopting dynamic prompt optimization.
- **Open-Weight Alternatives:** In response to the opacity of proprietary models, there is a marked shift toward evaluating and deploying open-weight models where version control is absolute.

5. Research Challenges and Limitations

Despite advances in evaluation frameworks, several severe challenges impede the comprehensive tracking of performance drift.

First and foremost is the "black-box" nature of commercial LLMs. Model providers rarely publish the specific weights, exact training data additions, or the exact nature of the RLHF (Reinforcement Learning from Human Feedback) tuning applied during updates. This opacity restricts researchers to observational studies. When a model's performance on a benchmark drops, it is nearly impossible to definitively diagnose whether the cause is catastrophic forgetting, changes in the context window mechanics, or an overtuned safety classifier.

Additionally, evaluating generative AI inherently lacks automated ground truth for complex, open-ended tasks. While assessing code compilability (Tian et al., 2023) or math accuracy (Chen et al., 2024) is relatively deterministic, assessing shifts in "helpfulness" or "tone" requires expensive human annotation or reliance on LLM-as-a-judge frameworks, which are themselves subject to the same behavioral drift.

6. Research Gaps and Future Directions

The current state of the literature points to several critical gaps that must be addressed to mature the field of LLM operations.

- 1. Longitudinal Multi-turn Evaluation:** While single-turn benchmarks are widely used, tools for measuring continuous context decay in real-world, multi-turn agentic workflows remain nascent. Building on the equilibria models proposed by Dongre et al. (2025), future research should establish standardized metrics for conversational persistence across model versions.
- 2. Granular Taxonomy of Drift:** The community lacks a standardized taxonomy to differentiate between types of drift (e.g., semantic drift, formatting drift, alignment tax, and capability regression). Developing such a taxonomy will allow for more targeted mitigation strategies.
- 3. Predictive Modeling of Updates:** Rather than reacting to breakages post-update, future research should explore predictive techniques that can estimate how fine-tuning on specific safety datasets will ripple through a model's general capabilities, aiding in the creation of reliable misinformation mitigation strategies without compromising underlying utility (Pelrine et al., 2023).

7. Conclusion

As Large Language Models increasingly act as foundational infrastructure for global software ecosystems, the silent alteration of their underlying behavior presents a critical

vulnerability. The phenomenon of performance drift—evidenced by fluctuating capabilities in logical reasoning, coding, and instruction following—demonstrates that current LLMs are not static artifacts but highly volatile services. Moving forward, both industry and academia must prioritize the development of continuous, automated evaluation frameworks. Until absolute transparency is achieved in how commercial models are updated, rigorous empirical tracking remains the only defense against the unpredictable trajectories of artificial intelligence behavior.

References

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